Dhruv Sarkar

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Research Interests

Networked AI Agents, Sequential Decision Making and Interactive Learning; Test-Time Alignment of LLMs; Generative Models; Theoretical Foundations of Machine Learning; Stochastic Optimization; Computational Optimal Transport.

Education

Massachusetts Institute of Technology, Cambridge, MA, USA **2026- Present**
S.M. and PhD, MIT Media Lab
Thesis advisor: Prof. Ramesh Raskar

Indian Institute of Technology Kharagpur, India **2021 - 2026**
B.Tech. (Honours) + M.Tech., Department of Computer Science and Engineering
Thesis advisors: Prof. Abhishek Sinha and Prof. Sayak Ray Chowdhury

Publications

Conference Publications

1. **Optimal Anytime Algorithms for Online Convex Optimization with Adversarial Constraints**
   Dhruv Sarkar, Abhishek Sinha
   The 43rd International Conference on Machine Learning (ICML 2026)
2. **Improved Algorithms for Nash Welfare in Linear Bandits**
   Dhruv Sarkar, Nishant Pandey, Sayak Ray Chowdhury
   The 43rd International Conference on Machine Learning (ICML 2026)
3. **Projection-free Algorithms for Online Convex Optimization with Adversarial Constraints**
   Dhruv Sarkar, Aprameyo Chakrabartty, Subhamon Supantha, Palash Dey, Abhishek Sinha
   The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)
4. **Relation-Aware Slicing in Cross-Domain Alignment**
   Dhruv Sarkar, Aprameyo Chakrabartty, Anish Chakrabarty, Swagatam Das
   The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)
5. **Revisiting Social Welfare in Bandits: UCB is (Nearly) All You Need**
   Dhruv Sarkar, Nishant Pandey, Sayak Ray Chowdhury
   The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)
6. **DP-NCB: Privacy Preserving Fair Bandits**
   Dhruv Sarkar, Nishant Pandey, Sayak Ray Chowdhury
   The 40th Annual AAAI Conference on Artificial Intelligence (AAAI 2026)

Preprints and Workshops

1. **Non-Expansive Two-Time-Scale Stochastic Approximation: A Fixed-Schedule One-Quarter Barrier and Bias-Corrected Acceleration**
   Dhruv Sarkar, Vaneet Aggarwal
   Preprint
2. **High-Probability PL-SGD with Markovian Noise: Optimal Mixing and Tail Dependence**
   Dhruv Sarkar, Aprameyo Chakrabartty, Vaneet Aggarwal
   Preprint
3. **Nonlinear Two-Time-Scale Stochastic Approximation: A Sharp Phase Transition and How to Beat It**
   Dhruv Sarkar, Vaneet Aggarwal
   Preprint
4. **Price of Fairness in Bandits: A Tight Minimax Characterization**

Dhruv Sarkar, Soumyadeep Dutta, Sayak Ray Chowdhury
Under submission at SODA 2027

5. **A Geometric Approach to Constrained Online Learning**
Dhruv Sarkar, Abhishek Sinha
Preliminary version accepted at NeurIPS 2026 Second Workshop on MLxOR

6. **Constrained Contextual Bandits with Adversarial Contexts**
Dhruv Sarkar, Abhishek Sinha
Preprint

7. **Online Learning for Approximately-Convex Functions with Long-Term Adversarial Constraints**
Dhruv Sarkar, Samrat Mukhopadhyay, Abhishek Sinha
Under submission at NeurIPS 2026

Research and Professional Experience

Mohamed bin Zayed University of Artificial Intelligence **May 2026 – July 2026**
Visiting Researcher, hosted by Prof. Junpei Komiyama, Department of Machine Learning

Honors, Awards, and Grants

- **AAAI-26 Student Scholarship & Volunteer Award**: awarded a $1250 travel scholarship, complimentary registration and selection for the AAAI Student Volunteer Program at AAAI-26, Singapore.
- **AISTATS 2026 Registration Fee Waiver**: awarded complimentary registration by the International Conference on Artificial Intelligence and Statistics.
- Secured **rank 309** in the **Joint Entrance Examination 2021** among approximately 1 million candidates.
- Secured **rank 38** in the **ISI BStat entrance test 2021** organized by Indian Statistical Institute, Kolkata.
- Awarded the **Kishore Vaigyanik Protsahan Yojana (KVPY)** Fellowship by the Department of Science and Technology, Government of India; qualified in Grades 11 and 12 and ranked among the top 1000 awardees from approximately **100K** candidates.
- Ranked **2nd** in the **SUMMIT math camp** for gifted high school students organized by Indian Statistical Institute, Kolkata.
- Ranked in the national **top 1%** in the **Indian Olympiad Qualifier in Physics 2020–21** and **Indian Olympiad Qualifier in Chemistry 2020–21**.
- Awarded the **JBNSTS Junior Scholarship 2019** among high school students across West Bengal.

Teaching and Professional Service

- Teaching Assistant, **Statistical Learning Theory (CS60018)**, IIT Kharagpur, Autumn 2025.
- Teaching Assistant, **Programming and Data Structures (CS10003)**, IIT Kharagpur, Spring 2026.
- Reviewer for **AAAI 2026, 2027**.

Research Mentoring

- **Soumyadeep Dutta** Mentored research on fair and welfare-aware bandits, resulting in a co-authored preprint on the price of fairness in bandits. Currently an undergraduate at IIT Kharagpur.
- **Nishant Pandey** Mentored research on fair and welfare-aware bandits, resulting in papers at ICML, AISTATS, and AAAI. Currently at Google.

Selected Coursework

Mathematics: Probability & Statistics (A+), Advanced Calculus (A+), Linear Algebra (A+), Discrete Structures (A+), Computational Number Theory (A).
Machine Learning and Theory: Machine Learning (A), Advanced Machine Learning (A), Linear Algebra for AI and ML (A), Game Theory (A), Information Theory (A).

References

Prof. Ramesh Raskar
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Indian Institute of Technology Kanpur
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Prof. Vaneet Aggarwal
Edwardson School of Industrial Engineering
Purdue University
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Prof. Abhishek Sinha
School of Technology and Computer Science
Tata Institute of Fundamental Research
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Prof. Swagatam Das
Electronics and Communication Sciences Unit
Indian Statistical Institute Kolkata
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